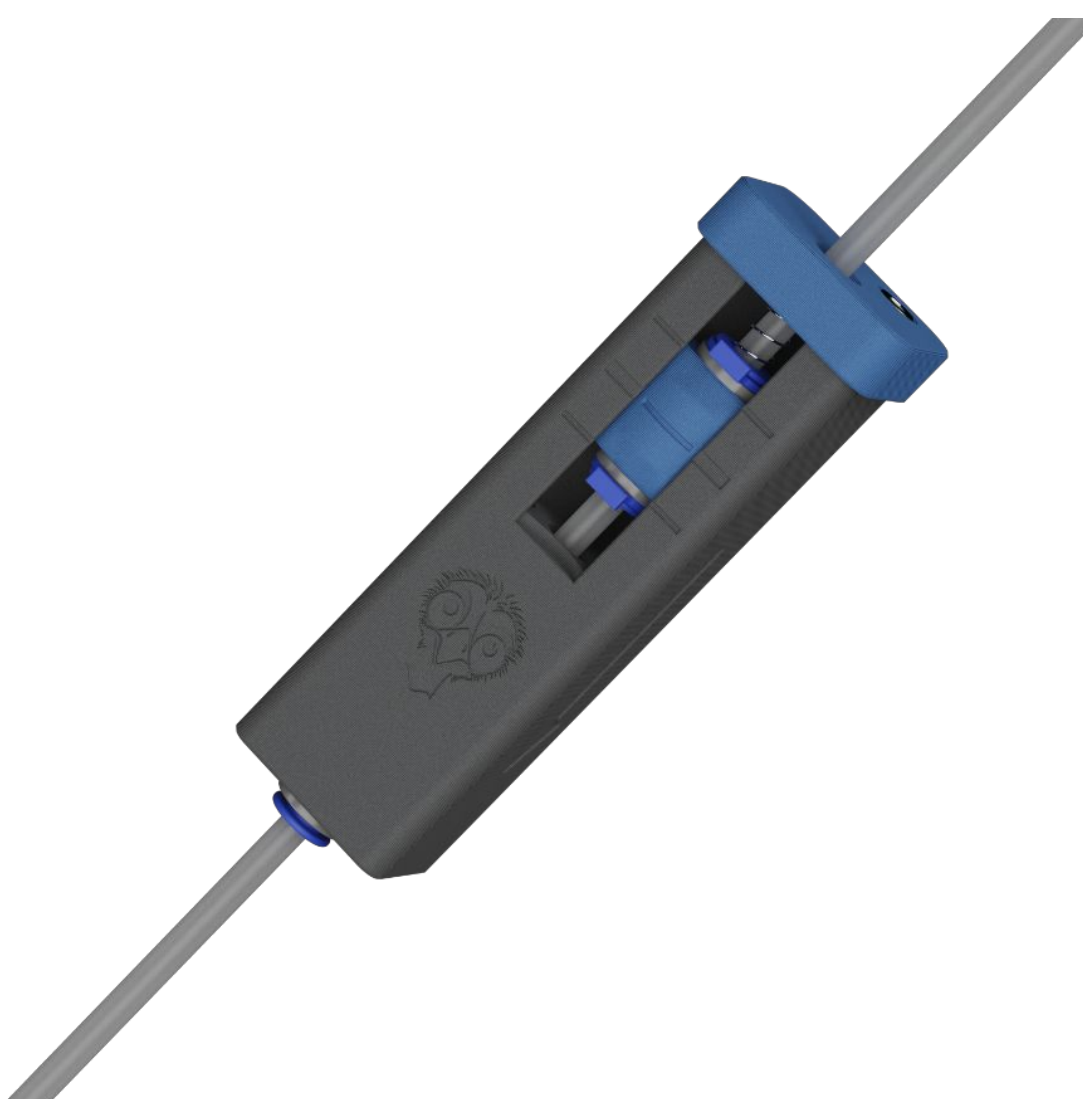




# EMUSync Housing for Proportional Sync Feedback Sensor

EMU - The Expandable Multi material Unit

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VERSION 1.02 - 2026-08-12

To have the best chance of success, the following print settings are highly recommended.

### 3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

### INFILL TYPE

Gyroid or Cubic

### MATERIAL

**ABS/ASA.** Glass and CF filled material are compatible.

### INFILL PERCENTAGE

Recommended: **15%-40%**

### LAYER HEIGHT

Mandatory: **0.2mm, including first layer!**

### WALL COUNT

Recommended: **4**

**One Wall Top and Bottom surface**

### EXTRUSION WIDTH

**0.4mm, Arachne slicing engine.**

Infill can be printed wider (0.45/0.48) to increase layer bonding.

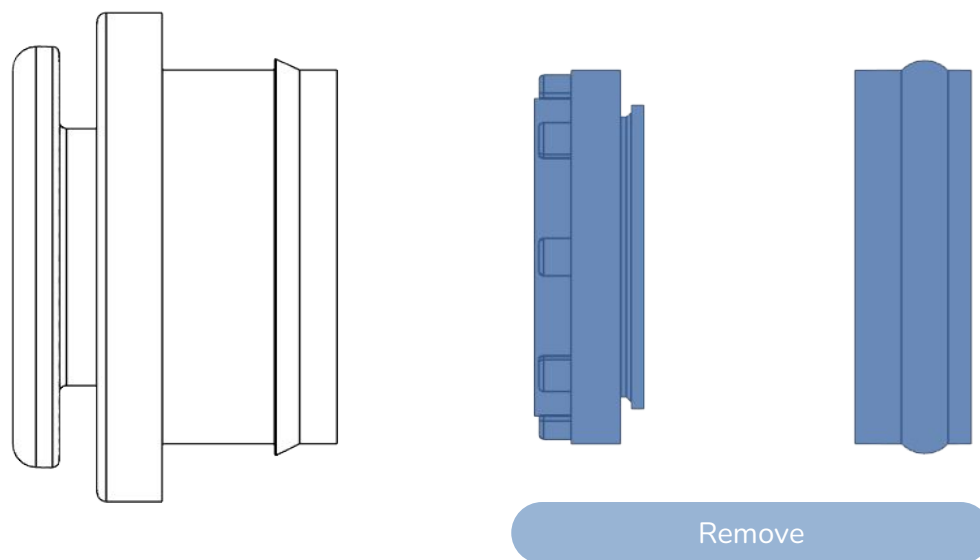
### SOLID TOP/BOTTOM LAYERS

Recommended: **5**

### SHRINKAGE COMPENSATION

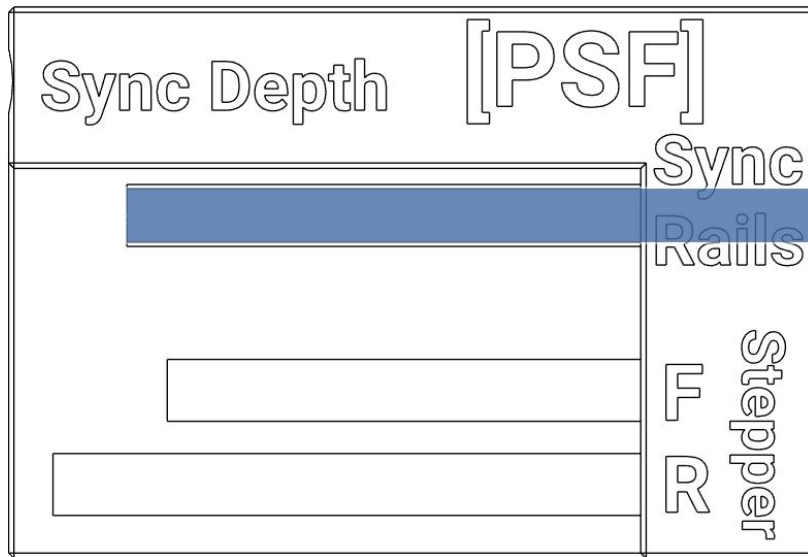
**Parts not compensated for shrinkage.**

Calibrate shrinkage compensation in the slicer before printing.



**CAUTION:**

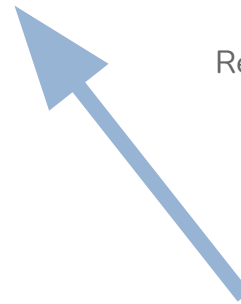
Whenever we are using ECAS in this manual, the seal retainer and the seal should be removed



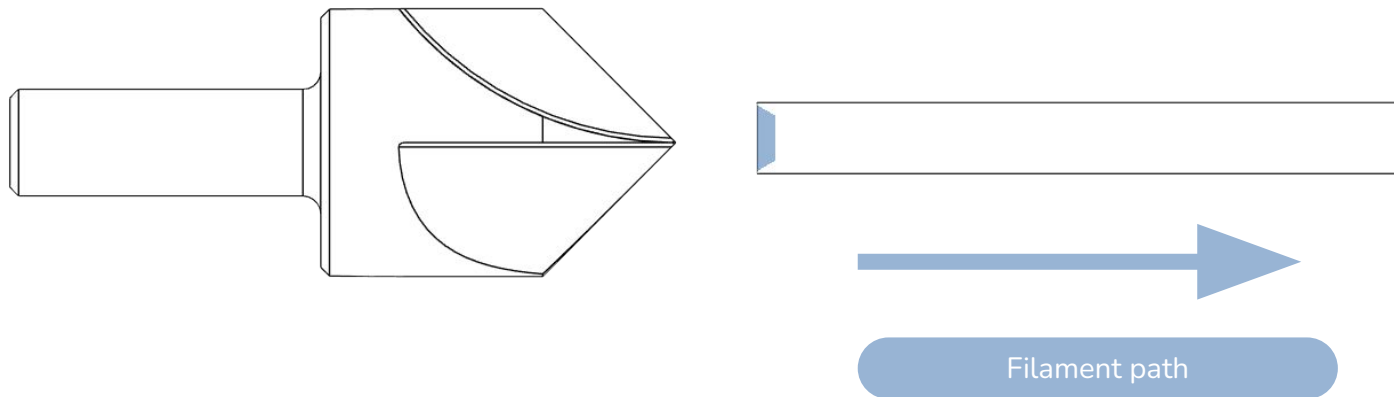
PTFE Cutter and Length Tool[PSF].stl

Insert a **4mm OD, 2.5mm ID** piece of bowden tube in the cutting jig. Cut flush with the body using a blade.

Repeat for the second rail, creating **two cut to size "rails"**.



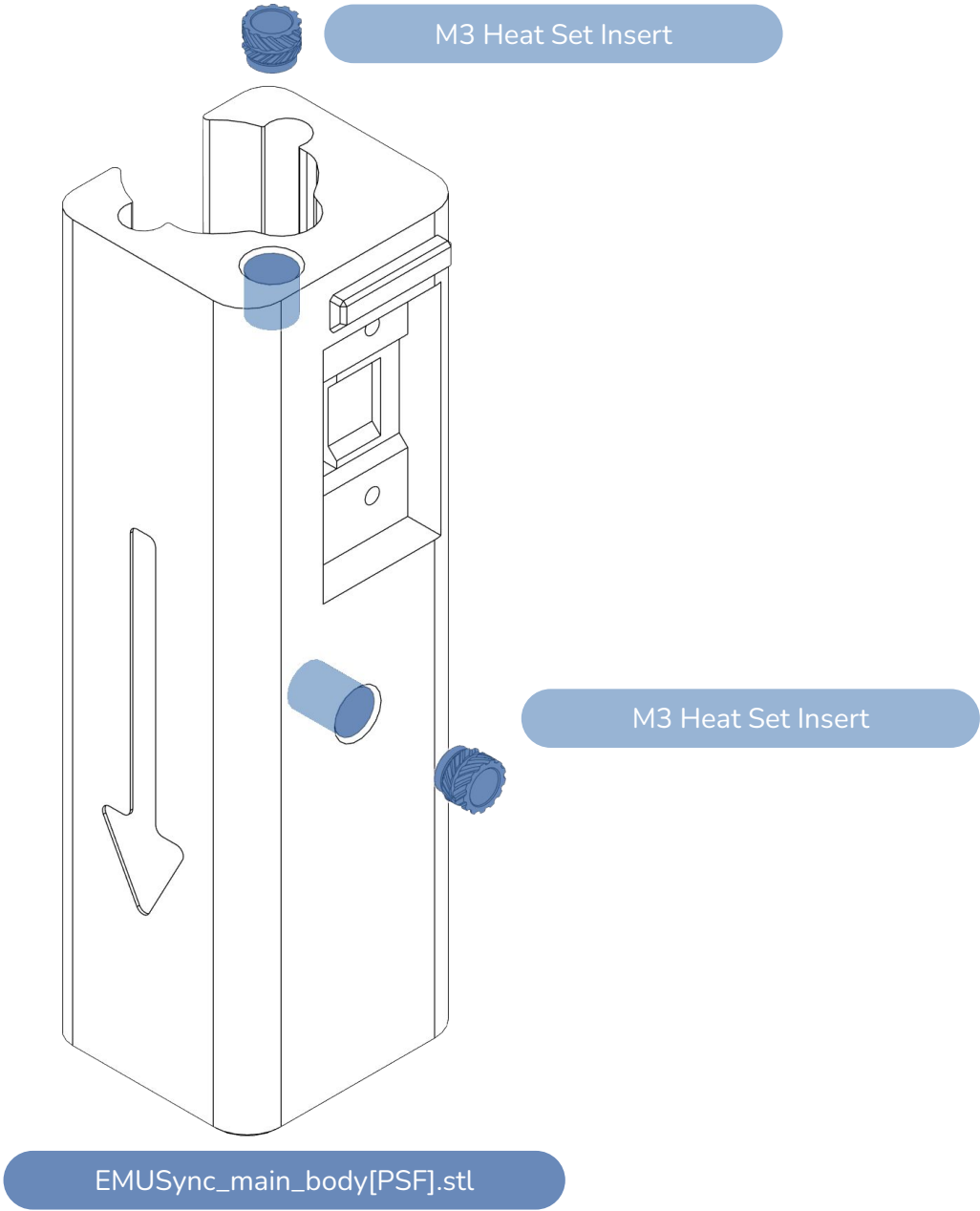
Cut here

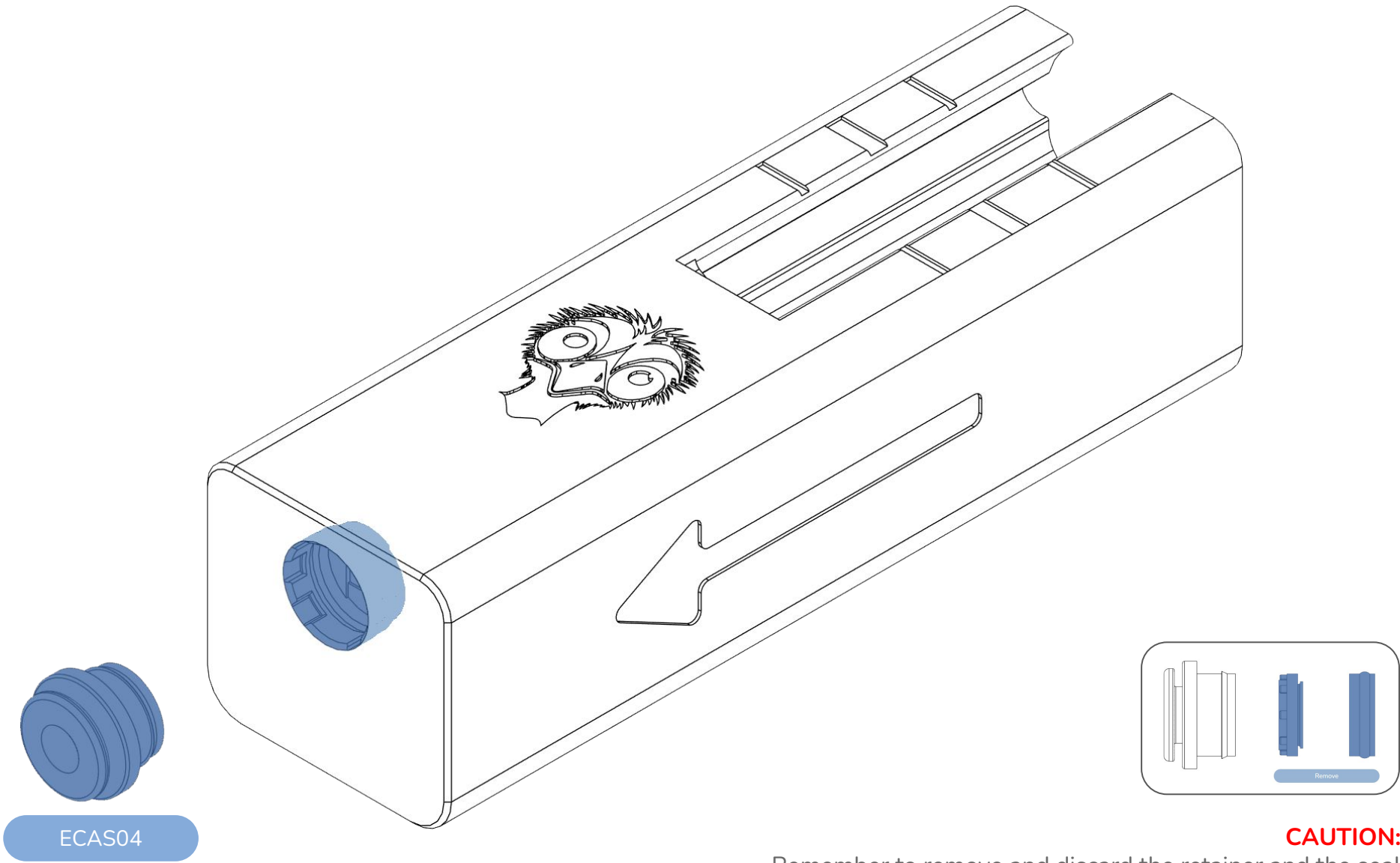


### CAUTION:

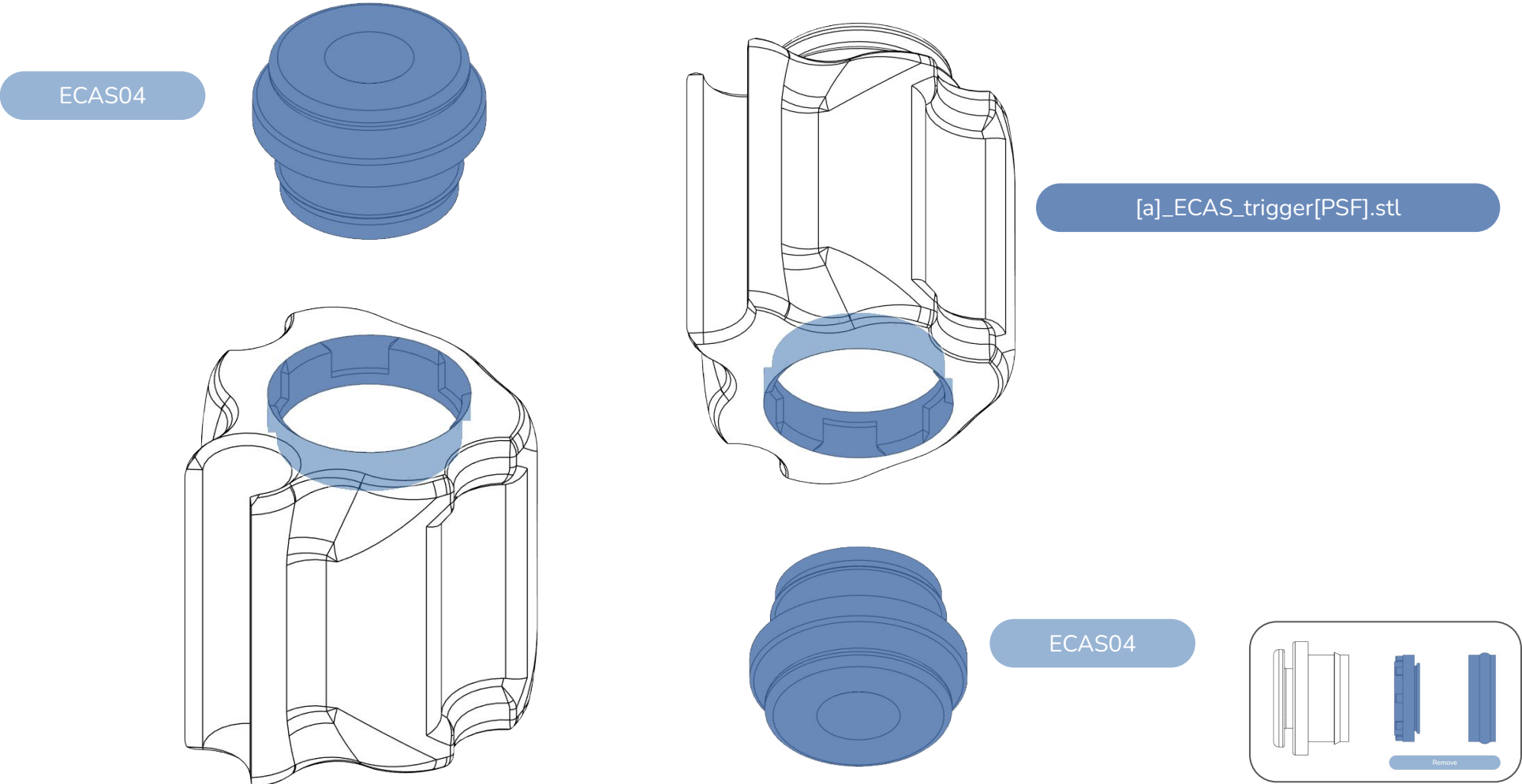
To ensure smooth feeding of the filament, it is recommended that the **inlet side of the PTFE tubes are slightly chamfered**. This can be done using a countersinking tool or a 4mm drill bit.

For the PSF, this means **the side of the tube that is inserted in the combiner exit** and the **side of the tube that is inserted in the EMUSync exit**.





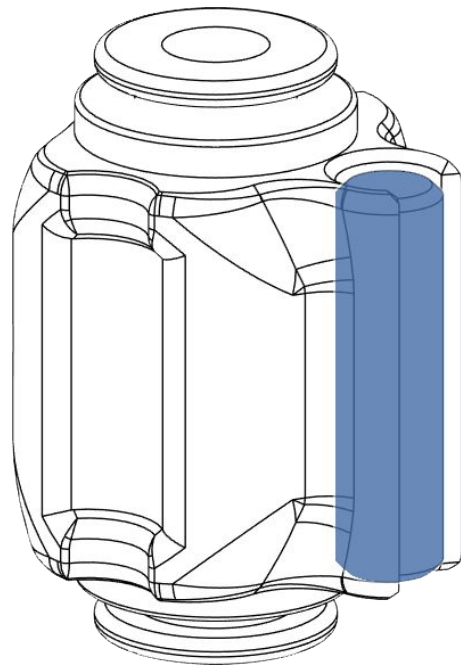
# INSERT ECAS FITTING IN THE SHUTTLE



**CAUTION:**

Remember to remove and discard the retainer and the seal

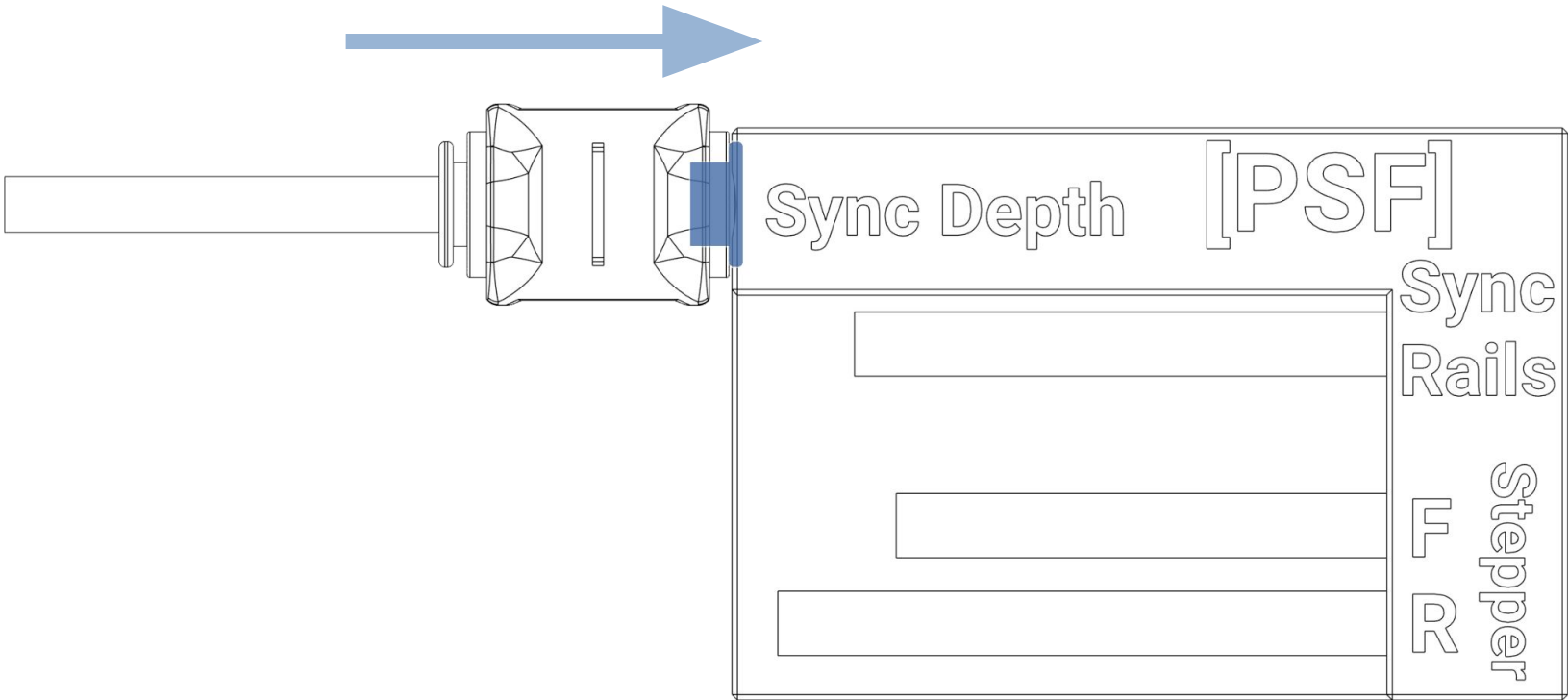




Magnet 4x15mm N35

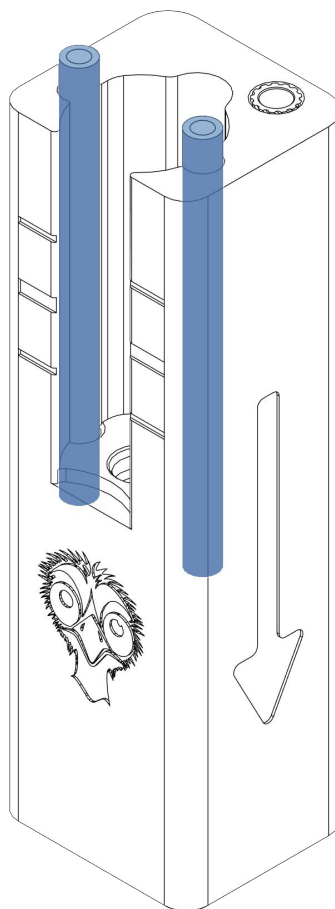
The orientation of the magnet doesn't matter, the software will auto calibrate no matter the orientation.

# INSERT THE BOWDEN TUBE IN THE SHUTTLE



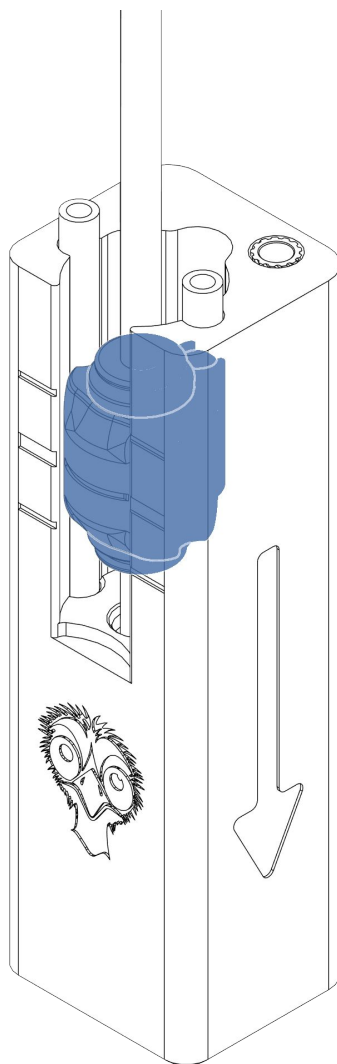
Use the cutting jig to keep the **right side ECAS fitting compressed**. While keeping it compressed, slide the PTFE tube in from the left side.

Push through until the tube bottoms out in the jig.



Insert the two PTFE rails into position. Push from the top and slide down **until they have bottomed out.**

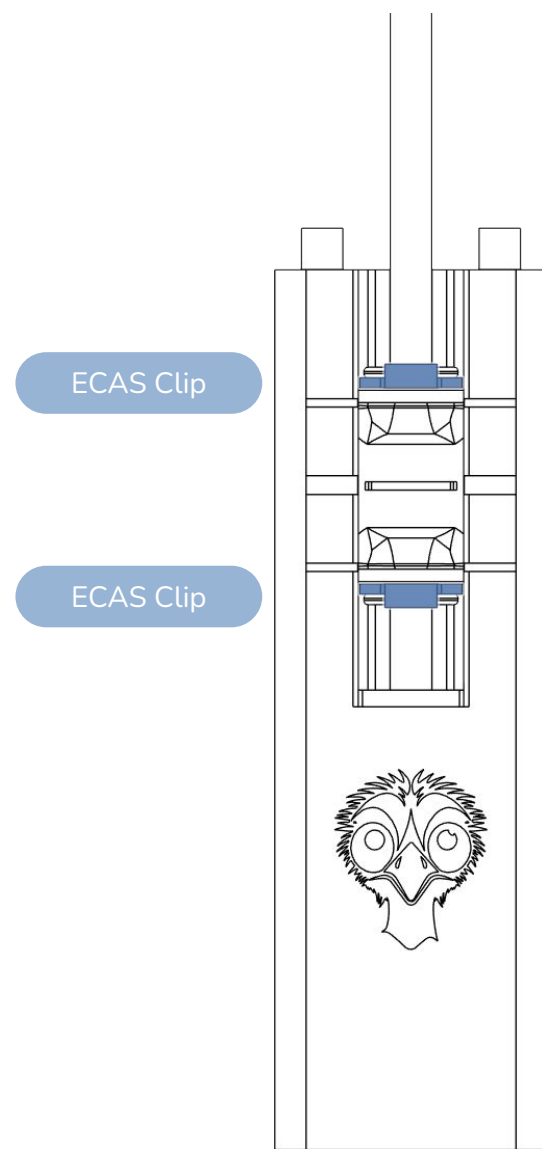
Try to not bend the tubes when inserting!



Slide the shuttle into position between the two PTFE rails you just placed.

## SECURE THE TUBE WITH THE ECAS CLIPS

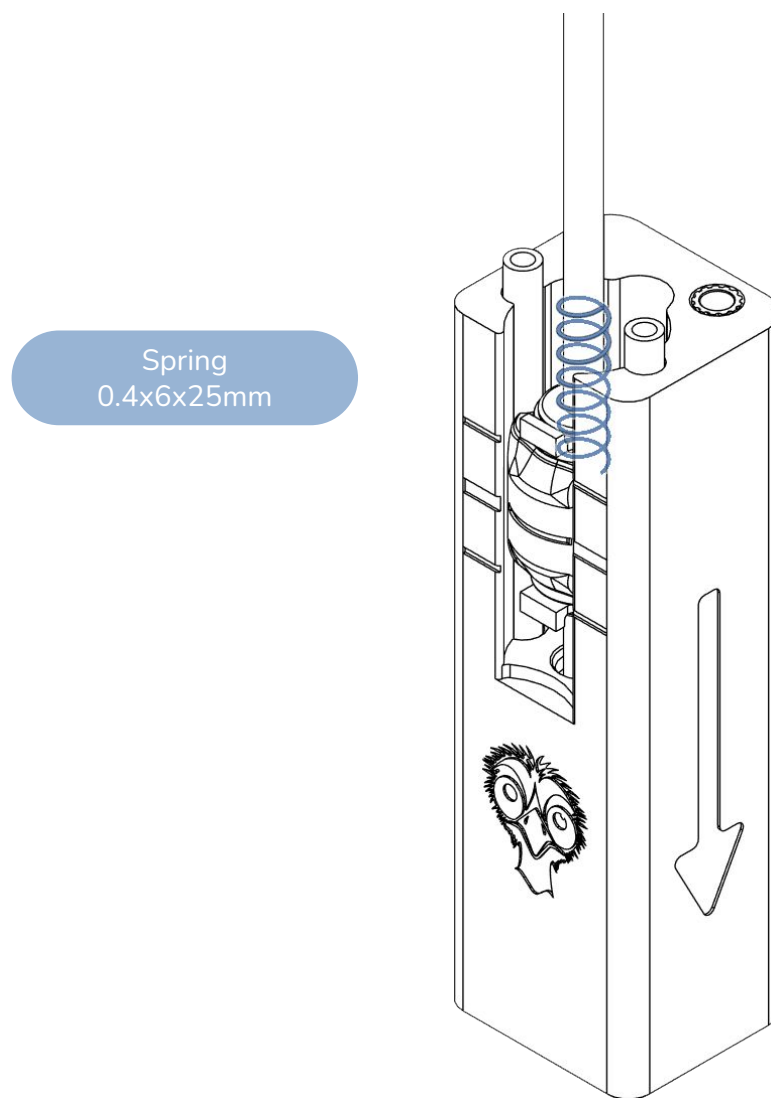
WWW.DWTAS.NET



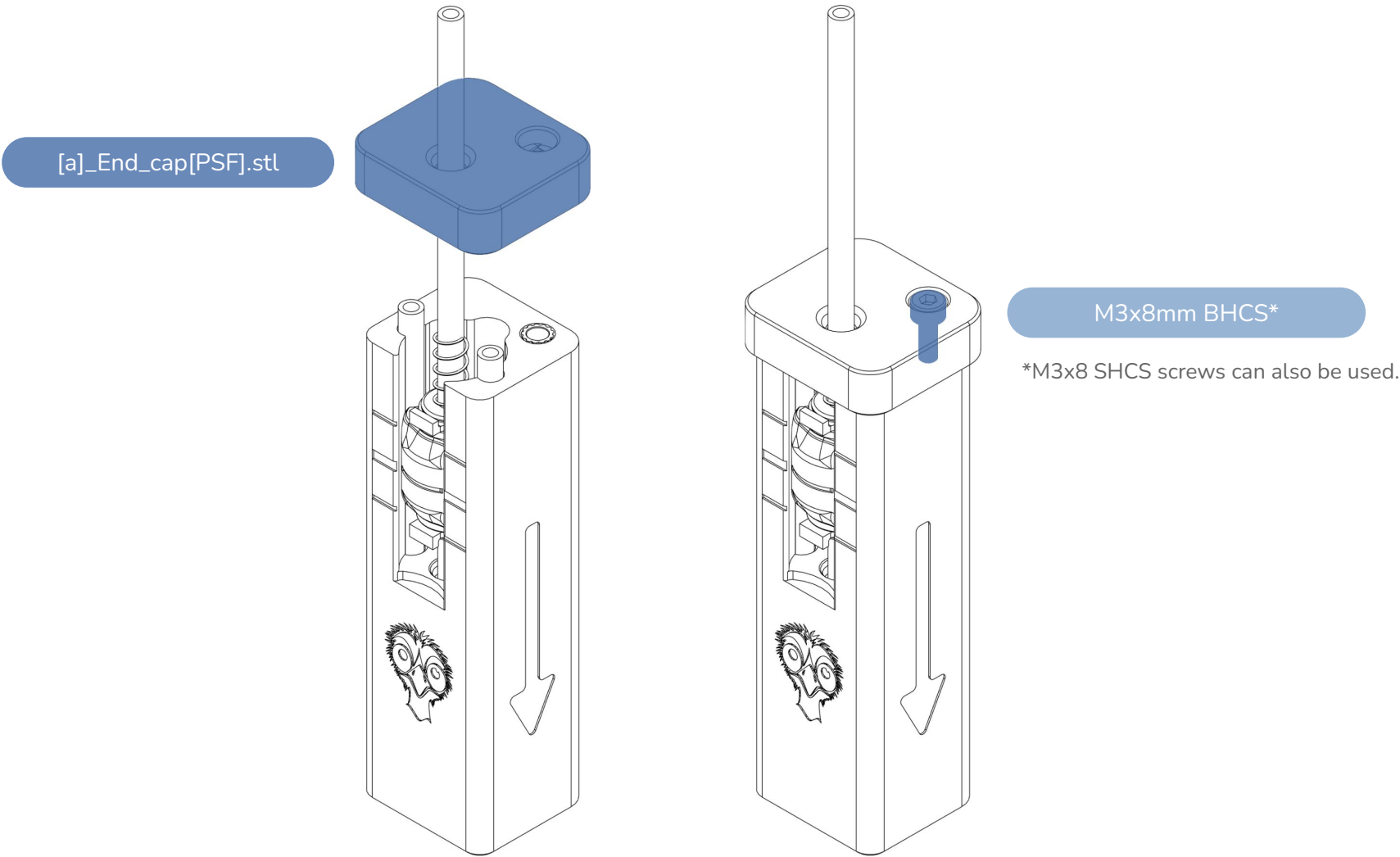
Place the ECAS clips to secure the tube in place. Be mindful of their orientation!

## PLACE THE COMPRESSION SPRING

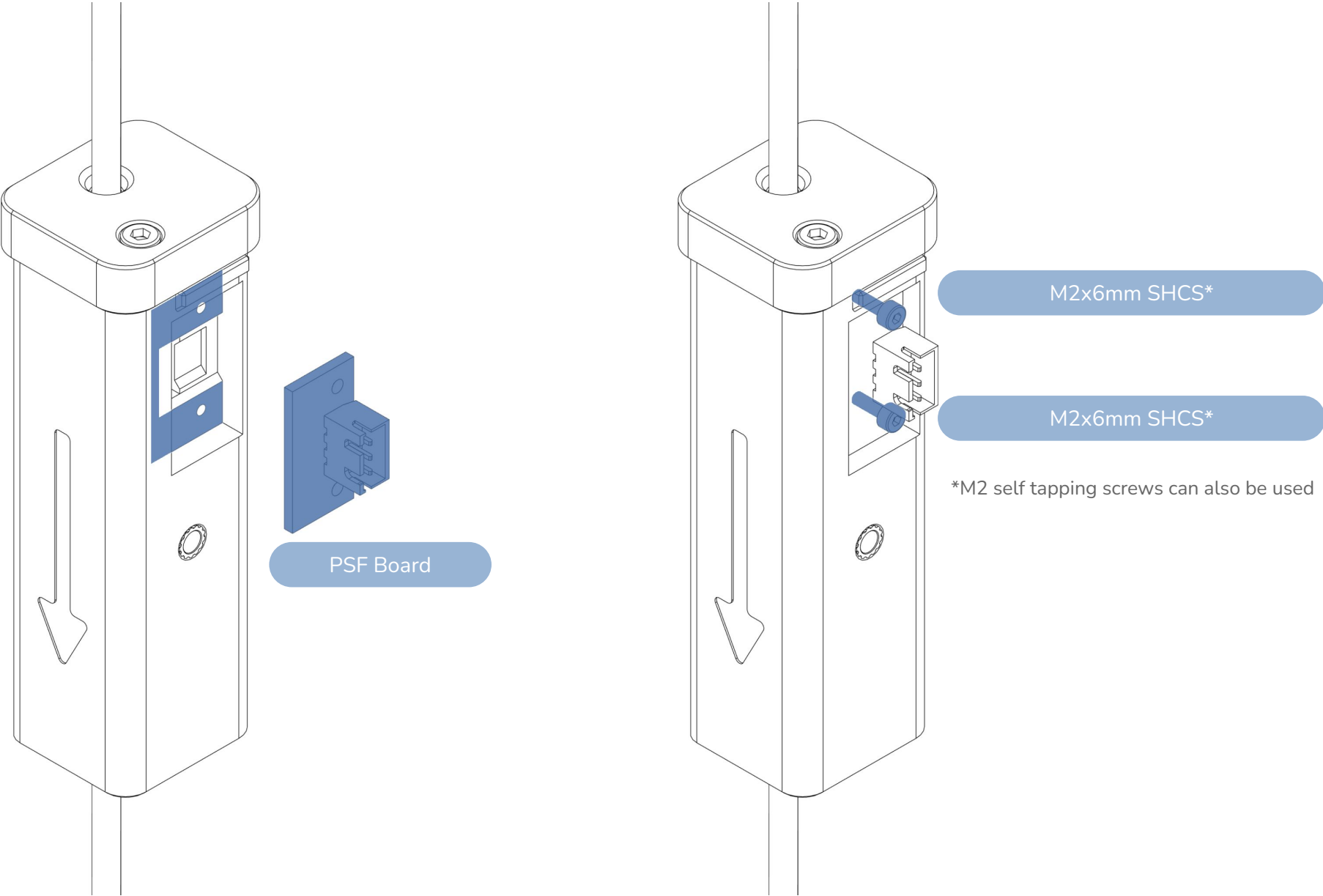
WWW.DWTAS.NET



# PLACE AND SECURE THE END CAP

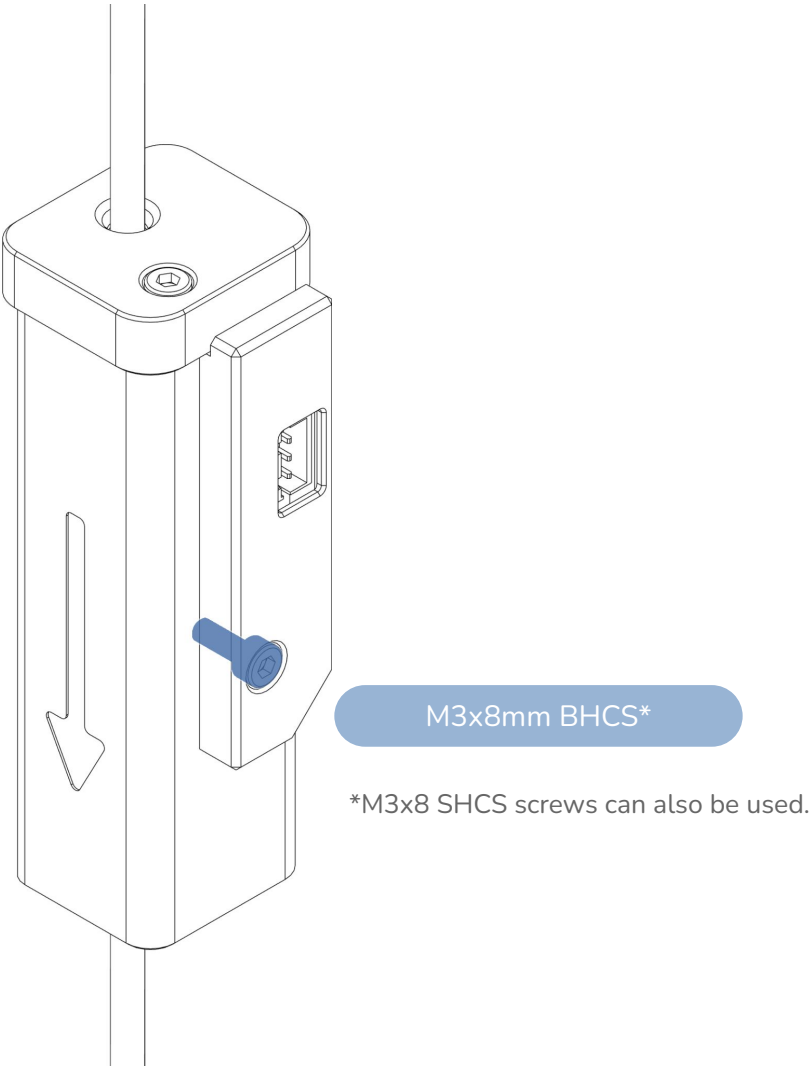
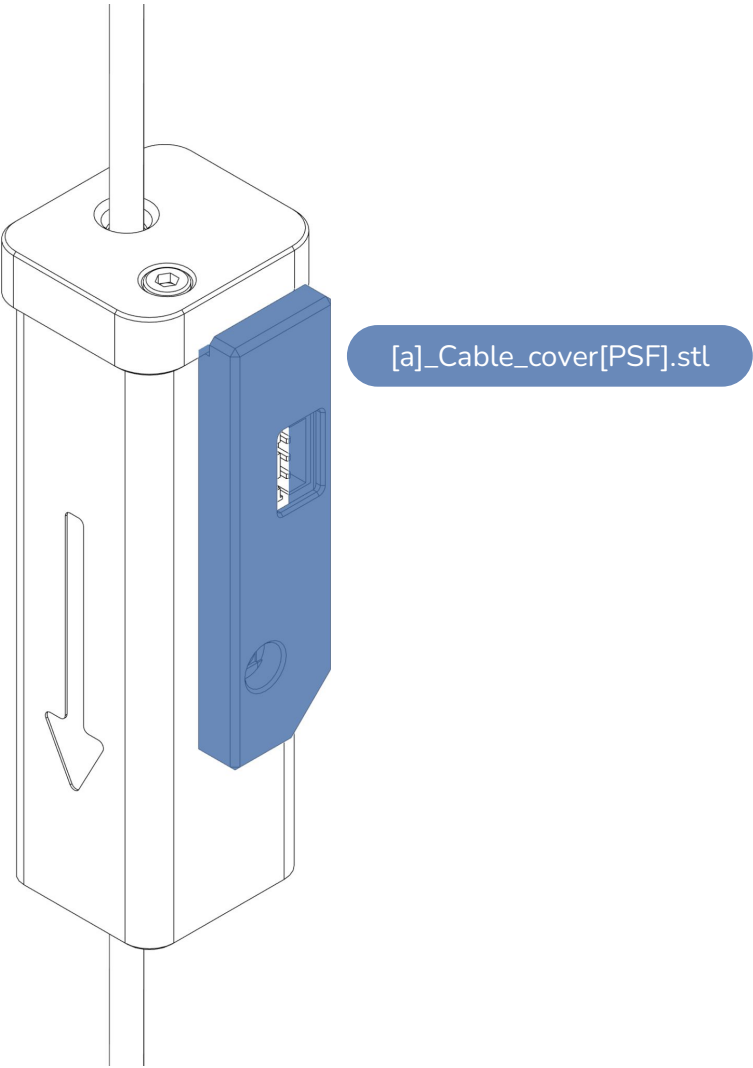


INSERT THE PSF BOARD

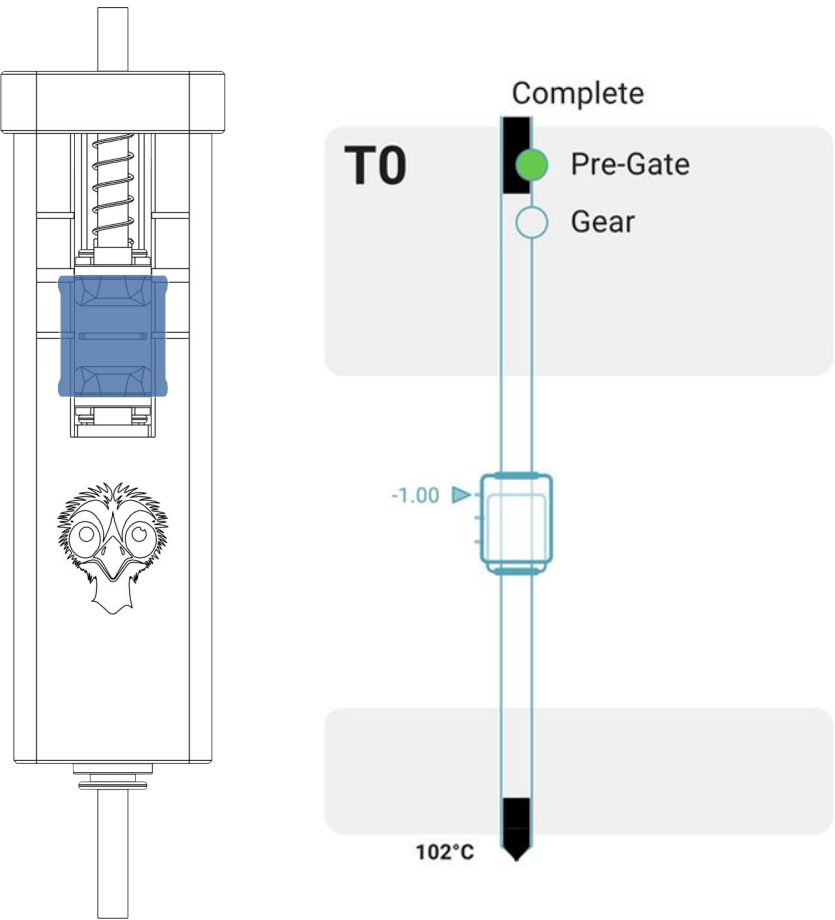




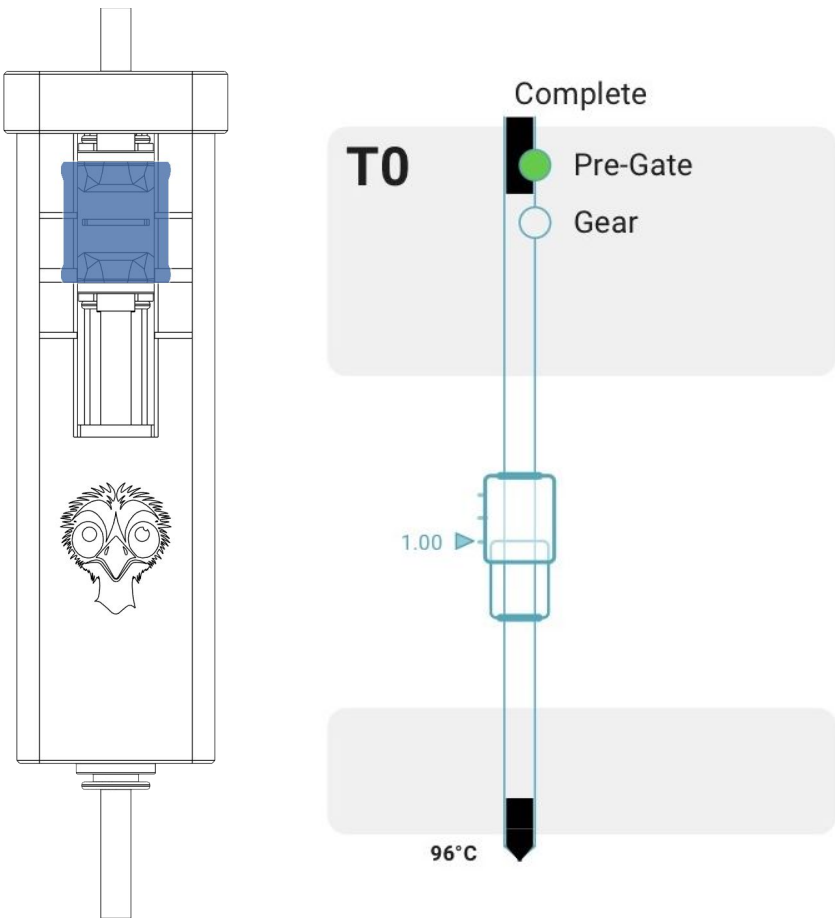
# PLACE AND SECURE THE CABLE COVER



Tension



Compression



THANK YOU

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EMU Github

<https://github.com/DW-Tas/EMU>



See my other projects

<https://dwtas.net/>



Join us on Discord !

<https://discord.gg/WuxGVTzu5p>

